

1 – EXERCISES ON VARIABLES

Problem 1

Each of the following pieces of code contains one or more error(s). Try to find them by hand.

You can check your answers by copy-pasting the code on <https://godbolt.org> and looking at the compiler's output. Try and get the code to compile without any error.

Listing 1:

```
1 int main() {  
2     int a_variable = " a variable ";  
3     return 0;  
4 }
```

Listing 2:

```
1 int main() {  
2     a = 0;  
3     b = 5;  
4     return a + b;  
5 }
```

Listing 3:

```
1 int main() {  
2     {  
3         int a = 5;  
4     }  
5     int b = 9;  
6     return a + b;  
7 }
```

Problem 2

A mixing engineer adjusts three tracks and sums them to produce a master level. What does `main` return? Try to find the answer without running the code on a computer.

Listing 4: Mixing desk

```
1  int main() {
2      int drums, bass, melody, master;
3
4      drums  = 80;
5      bass   = 70;
6      melody = 60;
7
8      drums  = drums + 10;
9      bass   = bass  * 2 - drums;
10     melody = melody + bass / 4;
11
12     master = drums + bass + melody;
13     drums  = master / 3;
14
15     return master - drums;
16 }
```

Problem 3

You are writing a simple image filter that brightens a pixel. A pixel has three colour channels stored as integers: `r`, `g`, `b` (each between 0 and 255).

Write a complete `main` function that:

1. Declares and initialises `r = 80`, `g = 120`, `b = 40`.
2. Declares an `int` variable `boost` and sets it to 30.
3. Adds `boost` to each of `r`, `g`, and `b` (reuse the same variables).
4. Declares an `int` variable `brightness` equal to the *average* of the three brightened channels (use parentheses to get the order of operations right!).
5. Returns `brightness`.